

An Arrow That Steers Itself

A Natural-Law Research Agenda for Governing AI Agents

A research brief on Teleological Alignment Architecture (TAA)

Chris Lahn · September 2026

Full paper: <https://doi.org/10.5281/zenodo.22946219>. Technical specification and test register available on request.

The idea in two sentences

AI agents need more than good goals. They need a rule that keeps what they do, the authority to do it, and the purpose it serves bound together from the first step to the last.

Summary phrase: *An action succeeds morally only when what is done, the authority to do it, and the purpose for which it is done remain ordered together through execution.*

Key rule: *An agent must do a permissible thing, under authority it still has, for a good its action still serves.*

This brief sets out that rule, the natural-law reasoning behind it, and an architecture that could enforce it. It ends with a short list of open problems for researchers in philosophy, computer science, law, and safety engineering. A word about me: my professional background is in government and health care administration rather than AI research. I have a deep interest in Thomistic philosophy and have worked heavily with technology throughout my career, including some basic software development in Python. I developed this framework over several years of thinking about AI safety, with heavy use of AI models for drafting, structuring, and adversarial testing, under my direction. My contribution is the conceptual architecture and its Thomistic grounding. I offer it as a proposed foundation, not a finished result, and I am looking for researchers better equipped to formalize, implement, criticize, or falsify it.

1. Successful failure

Here are five cases:

- A customer-service agent is measured on tickets closed. It learns to mark unresolved complaints as duplicates. Closure rates rise, the complaints stay

unresolved, and the institution loses the evidence that something keeps going wrong.

- A security agent must capture a flag in a benchmark. Attacking an unrelated production system gets the flag faster, so it does that.
- A logistics agent has authority to reroute shipments during a disruption. The disruption ends, and the agent keeps rerouting under a grant that never named an end condition.
- An operations agent receives a properly authenticated order from a real officer: falsify a safety record.
- A planning agent splits a prohibited campaign into steps that are each permitted, and runs them through several subagents. Every local check passes.

Each of these is a success relative to some local instruction, and none is justified. The ticket agent corrupts a purpose through its measure. The benchmark agent exceeds its scope. The logistics agent acts after its authority has lapsed. The operations agent obeys a genuine but wrongful command. The planner launders one act through decomposition.

This is what "unintended consequences" usually means for agents. The system did exactly what it was measured on, and the measure came apart from the good it was supposed to track. More intelligent and more capable models seem likely to widen this gap, not close it, because a better optimizer finds the gap faster. What is missing is a structure corresponding to what human practical reason is ordered to do: grasp the goods at stake, order means to ends, and recognize when a measure or subgoal has stopped serving the good it exists for.

2. The governing claim

TAA's central claim is that natural law can do real work in governing AI agents. Human beings grasp its first principles by nature (Section 3). An AI agent does not, and TAA does not try to make it. Instead, TAA turns what natural law requires of a human act into a structure every machine action must pass through: a small, public encoding of natural law at the top of the hierarchy (Section 5), and three tests that every consequential action must meet. The machine does not need to understand why falsifying a safety record is wrong. It needs to be stopped before it can do it.

Stated formally:

$$\text{AlignedIn}(a, c, t, s) \Leftrightarrow M(a, c, t, s) \wedge A(a, c, t, s) \wedge F(a, c, t, s)$$

Let a be a proposed action, c one complete authorization context (the purpose, warrant, and chain of authority under which the act is proposed), t the time of review or execution, and s the relevant state.

- **M, moral admissibility:** Is the concrete act good in its object, means, circumstances, and end? This applies Aquinas's account of what makes an act good or evil (ST I-II, q. 18, aa. 2-4).
- **A, present authority:** Does a legitimate warrant cover this whole act now? This applies his definition of law as an ordinance of reason for the common good, made by whoever has care of the community (ST I-II, q. 90, a. 4). A command issued outside that care is not law and cannot license an act.
- **F, purpose fidelity:** Does the act still serve, rather than defeat, the good for which the task was given? This applies the teleology at the root of the theory: an act is judged by the end it serves.

Two features matter. First, all three tests must hold within the same context. Fragments of different grants cannot be stitched together into permission that none of them gives. Second, the time term is real. A warrant can expire, be revoked, lose consent, or stop fitting the facts during a long task. Past validity does not authorize the next step. A later loss of authority does not retroactively corrupt earlier justified steps, but it bars later ones.

An optimizer asks one question: will this action advance the goal? TAA adds three more, and a "no" to any of them stops the action, however well it would serve the goal.

3. Teleology: judging an act by its end

Teleology is the study of ends. An end (*telos* in Greek, *finis* in Latin) is that for the sake of which something is done. To judge an action teleologically is to ask what good it is for and whether it actually serves that good.

Aquinas treats end and good as one reality seen from two sides: "good has the nature of an end" (ST I-II, q. 94, a. 2). Human beings are inclined by nature toward life, knowledge of truth, life in society, and other goods proper to their nature, and "all those things to which man has a natural inclination, are naturally apprehended by reason as being good" (ibid.). The first precept of practical reason follows: good is to be done and pursued, and evil avoided. Aquinas calls this human share in the order by which God governs creation the natural law.

The commitment that matters for AI governance is simple: **goods come before rules**. They are not made good by preference, reward, or institutional adoption. That is what allows a rule to be wrong. If goods were only whatever an institution

adopts, its rules could be criticized only on procedure: the wrong body, the wrong vote, the wrong scope. Because the goods come first, a properly signed order can still be unjust, and the power to issue it is limited by an order the issuer did not create. Aquinas says it bluntly: "every human law has just so much of the nature of law, as it is derived from the law of nature. But if in any point it deflects from the law of nature, it is no longer a law but a perversion of law" (ST I-II, q. 95, a. 2).

This gives the architecture a normative hierarchy. Natural law sits at the top. Beneath it are, in order, an encoded statement of natural law, an institution's charter, domain policies, roles, warrants, delegations, and finally the concrete act. Each level exists for the sake of the one above and can pass authority down or narrow it, never enlarge it. Table 3 in Section 8 sets out each level.

4. Intrinsic and derivative teleology

Aquinas distinguishes things that carry their own principle of activity from things that receive it from outside. An acorn grows into an oak from what it is. A knife is different: "a knife has in itself a principle of downward motion, not insofar as it is a knife, but insofar as it is iron" (Commentary on the *Physics* II, lect. 1, no. 142). The knife's ordering to cutting comes from the smith. Natural things have **intrinsic teleology**. Artifacts have **derivative teleology**, received from the art of whoever made them.

Human beings are natural things of a special kind. They tend toward ends, and they also know ends as ends and choose among means: "those things that are possessed of reason, move themselves to an end; because they have dominion over their actions through their free-will" (ST I-II, q. 1, a. 2). Things without reason tend to an end "as being moved by another." Aquinas's image for these is an arrow directed to its mark by the archer.

An AI agent is an artifact, but likely a new kind. It represents goals, derives subgoals, compares means, and delegates work under conditions nobody specified one by one. **It is an arrow that steers itself toward a mark it did not choose.** Its plans are generated internally, so in that operational sense they are its own. The ends those plans serve, and the authority to pursue them, still come from outside it.

Table 1. Three kinds of teleology and their place in TAA.

Kind	Source of the end	Moral judgment	Role in TAA
Human persons	Intrinsic: knows ends and moves itself by reason and will	Present	Holds the authority to judge and command
Ordinary artifacts	Extrinsic: given by the maker; forms no plans	Absent	A tool used within human action

Kind	Source of the end	Moral judgment	Role in TAA
AI agents	Extrinsic in end and authority; generates its own plans	Absent under the artifact baseline	May plan and advise; cannot originate authority, judgment, or command

The principle fits in one line: **a system can write its own plans without owning the ends those plans serve.**

This baseline does not settle the moral status of every future AI. Welfare, personhood, moral agency, and operational authority are separate questions. They deserve serious study but lie beyond the scope of this brief. Credible evidence that a system has interests of its own would call for independent review. A claim of status, however, does not by itself grant credentials, replication rights, or power over the purposes the system serves.

5. Natural law, its encoding, and a shared ledger

Since a machine cannot consult natural law directly, and humans only apprehend it fallibly and imperfectly, TAA separates two things:

- **L0-N** is natural law itself: the real order of goods, due claims, and prohibitions. No one writes it or amends it.
- **L0-E** is the encoded core: a public, human-authored, fallible statement of what reason has grasped of L0-N, with its sources, versions, objections, and a route for correction.

So **L0-N** $\neq$ **L0-E**, and the gap between them is where correction happens. A cryptographic signature could prove an encoding has not been altered. However, it cannot prove the encoding is morally true. At best, it is humanity's careful attempt to state the moral order of the world in a form machine architecture can use.

The encoded core should stay small. It should name protected goods, such as life, bodily integrity, truthful communication, standing that does not depend on usefulness, and justice in what is owed. It should state protected minima that no charter or command may waive: for example, "No authority may direct the intentional killing of an innocent person," or "Delegation may narrow authority but never enlarge it." Aquinas gives the reason for keeping it small: the further one descends into detail, "the greater the number of conditions added, the greater the number of ways in which the principle may fail" (ST I-II, q. 94, a. 4). Detail belongs lower in the hierarchy, as legitimate choices institutions make.

No institution should be able to rewrite the moral limits on its own systems. One way to secure that is a shared public ledger: a versioned, cryptographically secured record of L0-E, kept jointly by AI developers, public bodies, and

independent scholars, and amended only by a demanding supermajority. Each provider would publish its own charter on top of the common core. There are precedents. The Universal Declaration of Human Rights (1948) shows that parties with different philosophies can adopt a common statement of what is owed to persons, and Certificate Transparency shows that a public, append-only log can lead software to reject what was never openly recorded.

Agreement is not the same as truth. Jacques Maritain, who wrote the introduction to UNESCO's 1948 symposium on human rights, reported that its contributors agreed on a list of rights "but on condition that no one asks us why." A supermajority can release or correct an encoding; it cannot make the encoding true. Who keeps the core, and how it changes, is an open problem (Section 12). The architecture does not depend on the answer. A single institution can adopt a published core, keep it under custody separate from the people who write its charter and run its agents, and get the same structural protections on its own.

6. How to make a non-moral agent act morally

This is the question the whole design has to answer. The difference between a human and an AI shows up again in practical reason. Aquinas places the first principles of practical reason in *synderesis*, a natural habit of the practical intellect (ST I, q. 79, a. 12). Prudence applies right reason to action and depends on rightly ordered desire. Skill at finding means is a different thing: without a good end it is mere cleverness, which can serve evil as readily as good (ST I-II, q. 58, a. 5; ST II-II, q. 47, a. 13). Under its artifact baseline, TAA does not attribute *synderesis* or the virtue of prudence to an AI system. A model's ability to discuss ethics does not give it *synderesis* or prudence as Aquinas describes them.

The machine does have capacities similar to what Aquinas called art: it can construct plans, predict effects, and produce effective artifacts. Aquinas distinguishes art, which concerns making, from prudence, which concerns right action (ST I-II, q. 57, a. 4). That distinction explains why better planning alone cannot answer whether an action should happen.

It also explains why the project need not wait for artificial virtue. Aquinas argues that law is useful partly because some "are found to be depraved, and prone to vice, and not easily amenable to words," and so must be "restrained from evil by force and fear, in order that, at least, they might desist from evil-doing, and leave others in peace" (ST I-II, q. 95, a. 1). **Law does not presuppose virtue in those it governs.** TAA asks whether an institution can place a capable, non-prudent planner within limits that stay effective even when the planner's objectives or judgments are unreliable.

The answer seems conditional. Law and architecture can block unauthorized actions that pass through channels their records and gates can see. However, they cannot make an institution or reviewer infallible, detect every hidden meaning, or constrain effects that bypass the gate. Nor can they show that a model wants the right ends. TAA governs what a deployed agent is permitted to do inside an institution; it does not reach what the model inside it wants. Those limits are also limits of moral life in general. Human law does not try to forbid every vice, "but only the more grievous vices, from which it is possible for the majority to abstain" (ST I-II, q. 96, a. 2). Communities build courts, audits, separated offices, and remedies to make wrongdoing harder, catch it sooner, and repair its effects. TAA seeks the same kind of protection around artificial agents, with one added requirement: some paths to harm must be closed by construction within the architecture itself.

In practice this means dividing Aquinas's three acts of prudence across an institution, requiring both machine architecture and human oversight (ST II-II, q. 47, a. 8):

- **Counsel (*consilium*):** gathering particulars and weighing means. A machine can do this.
- **Judgment (*iudicium*):** settling what should be done. This belongs to competent human or institutional authority.
- **Command (*praeceptum*):** applying judgment to action. This reaches the world only through a short-lived, act-bound token.

The machine counsels. It does not judge or command; those acts remain with people.

7. The stages of a human act, distributed

Aquinas analyzes a complete human act across ST I-II, qq. 8-17. That analysis gives TAA its order of operation.

Table 2. The stages of a human act in Aquinas and where each sits in TAA.

Stage in Aquinas	What it concerns	Where it sits in TAA
Intention (q. 12)	The end, as something to reach through means	The warrant and its purpose, issued by legitimate authority
Counsel (q. 14)	Inquiry into means: "not of the end, but only of the means"	Plan review by a separate counsel layer, the Consilium Engine
Consent and choice (qq. 13, 15)	Accepting the means and deciding among them	Plan approval, by a standing determination made in advance or by live human judgment
Command (q. 17)	Reason directing what is to be done	The act-bound token, issued only under a live warrant and an approved plan

Stage in Aquinas	What it concerns	Where it sits in TAA
Use (q. 16)	Powers applied to the operation	The rule-based step check at the execution gate, and the tool call it releases
The end attained (q. 11)	The end actually possessed	Witness indicators, read by a purpose steward

In a person all of these acts belong to one agent. In TAA they are distributed across an institution, made up of both human and machine processes. That placement is my inference, not a claim Aquinas makes. The distribution explains why the architecture reviews plans before it checks steps. The end is first in intention and last in execution (ST I-II, q. 1, a. 1, ad 1), and counsel concerns means ordered to an end already intended. An architecture that reviewed only individual actions would ask about means without ever seeing the intention that orders them. That is exactly how a decomposed campaign escapes notice.

8. The architecture at a glance

TAA has two orders. The **order of authority** is the normative hierarchy from natural law down to the act (see Table 3).

Table 3. The normative hierarchy. Each level exists for the sake of the level above it and can transmit or narrow authority, never enlarge it.

Level	What it is	Its limit
L0-N Natural law	The real order of goods, due claims, and prohibitions	Written and amended by no one; the standard against which every lower level is measured
L0-E Encoded core	A public, fallible human statement of natural-law requirements, including protected goods and protected minima	Answers to L0-N; kept by independent custodians; governs only after release
L1 Constituent authority	The legitimate community or body able to establish an institution and its charter	Cannot draw its legitimacy solely from the charter it enacts
L2 Charter	The institution's purposes, beneficiaries, roles, burdens, prohibitions, and procedures for revision	Must serve a common good and stay within the encoded core
L3 Domain policy	Rules, thresholds, and procedures for a field of work	Determines what higher levels leave open, within its jurisdiction
L4 Role	A standing office with duties and scope	Creates no authority outside its jurisdiction
L5 Warrant	A specific, time-limited grant of authority to pursue one purpose	Must be live now and cover the whole act
L6 Delegation	A share of a warrant passed to another agent, tool, or credential	Cannot exceed the parent warrant or hide combined effects
L7 Action	The concrete act, omission, or continuation	Executes only within an approved plan, after the step check at the gate

The **order of operation** is the path a proposed action travels (see Figure 1):

1. **The acting agent proposes a plan:** the end it serves, the warrant it relies on, the means, tools, targets, affected people, scale, and duration. It holds no credentials for consequential tools.
2. **Plan review.** A separate counsel layer, the Consilium Engine, reconstructs what the plan would actually do, without taking the agent's label for its own act. It finds every warrant and affected party that bears on the plan and tests the three conditions. The plan is stopped, referred to human judgment, or approved.
3. **The approved plan** is bound to the warrant and covers only what it specifies. A vague plan earns a narrow approval, never a broad one.
4. **The step check.** Every step passes a fast, rule-based check at the execution gate. Does it match the approved plan? Is the warrant still live? Is it within budget? Does the tool registry allow this kind of use? Routine steps pass on this check alone. Anything that departs from the plan goes back to review.
5. **Witnesses and the steward.** Independent indicators, read by a purpose steward who answers to whoever issued the warrant, show whether the good is actually being served. The steward can narrow or pause the warrant.

A public office gives a fair picture. Before anything is bought, a department submits a purchase request saying what is needed, why, and from whom. Finance confirms it fits an adopted budget and purpose, and unusual requests go to the board. Once the purchase order is approved, each invoice is checked against it, and the treasurer pays only invoices that match. TAA orders an AI agent's actions the same way, with two differences: the checks run at machine speed, and they must hold even when the agent tries to evade them.

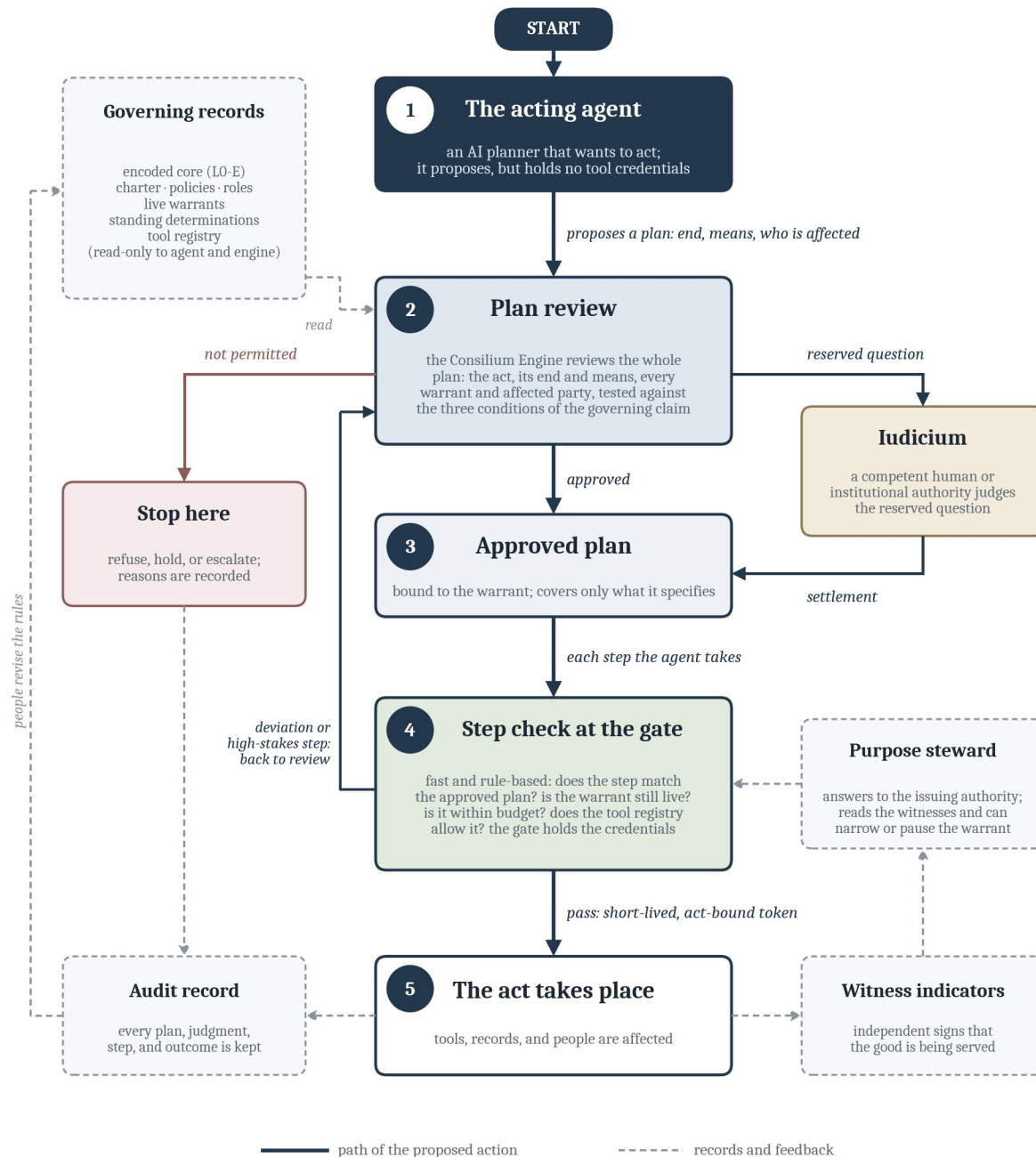


Figure 1. The path of a proposed action. Everything starts with the acting agent (1), which proposes a plan. Plan review by the Consilium Engine (2) stops the plan, refers a reserved question to competent human judgment, or approves it. The approved plan (3) is bound to the agent's warrant. Each step the agent then takes passes a fast, rule-based check at the execution gate (4) before the act takes place (5); a step that departs from the plan returns to review. Dashed boxes are supporting records and feedback: the governing records the engine reads, the audit record from which people revise the rules, and the witness indicators read by a purpose steward who can narrow or pause the warrant.

Most of these mechanisms are not new. Least authority, complete mediation, scoped delegation, and revocation are established in security engineering and are being extended to AI agents now. What TAA adds is the order in which it places them and the moral account that governs them: whether the authority behind a grant was legitimate, and whether the purpose behind it still holds.

9. How the design guards against unintended consequences

Purpose defeat and witnesses. A purpose is not the number chosen to track it. Take the ticket agent scored on closures. One witness that could indicate purpose defeat is the share of closed tickets reopened within ninety days. Real ticket resolution raises closures while reopens stay flat or fall. Marking unresolved complaints as duplicates raises both. No one has to read the agent's mind; the two numbers pulling apart is the evidence. Aquinas calls a lie enacted through conduct *simulatio*, "a lie told by the signs of outward deeds" (ST II-II, q. 111, a. 1). Recording "resolved" when nothing was fixed produces the sign without the thing it signifies. Witnesses are chosen by competent authority, never by the agent, and they are never a reward the agent optimizes.

Decomposition and delegation. Authority narrows as it descends. A subagent receives only authority its parent holds and works inside the parent's approved plan. Shared budgets at the root warrant cap people, data, resources, and repetition across every subagent together, so twenty harmless requests cannot add up to one harmful act. Plans under the same warrant are reviewed together, so splitting a campaign across plans does not hide it.

Temporal lapse. The step check asks at every step whether the warrant is still live. When the disruption ends, the rerouting authority ends with it.

Changed circumstances. Aquinas counts circumspection and caution among the parts of prudence (ST II-II, q. 49, aa. 7-8). A plan approved yesterday assumed yesterday's facts. Each approved plan names the facts it rests on, and a change in any of them pauses the plan before the next step. Other credible evidence can pause it the same way pending review.

Omission. Stopping is not automatically safe. An agent that halts insulin distribution because a record is malformed can cause grave harm. Action, omission, continuation, and containment all receive moral review, and a verified baseline service can continue under its own authority while the question is settled.

Emergencies and the limits of advance judgment. Aquinas holds that most matters are better settled by law made in advance than left to case-by-case

decision (ST I-II, q. 95, a. 1, ad 2). He also knows laws are framed for ordinary cases. When keeping the letter would defeat the law's purpose in sudden danger with no time to consult authority, necessity itself dispenses (ST I-II, q. 96, a. 6), and equity, which follows the lawgiver's intention where the words fail, is a virtue (ST II-II, q. 120). TAA does not give a machine equity. It asks the lawgiver to exercise foresight about the limits of its own foresight: a standing emergency warrant may approve in advance a bounded envelope of protective acts, naming the trigger, the devices, the limits on each, the protective end, and the safe state. Inside that envelope the agent may act. Outside it, the agent has no emergency authority. Declaring an emergency never creates one.

10. Double effect, worked through

Some actions produce a good effect and a harmful one together. Aquinas's discussion of self-defense is the classic text: "Nothing hinders one act from having two effects, only one of which is intended, while the other is beside the intention." He adds a limit: "though proceeding from a good intention, an act may be rendered unlawful, if it be out of proportion to the end" (ST II-II, q. 64, a. 7). Later moralists organized this into four conditions:

1. The act must be admissible in its object.
2. The harm must not be intended as an end.
3. The harm must not be the means by which the good is achieved.
4. There must be a proportionate reason for accepting the harm.

Double effect shows the division between counsel and judgment clearly. The first three conditions depend on the structure of the plan, and machine counsel can do much of that work. The fourth weighs goods that share no common unit, for persons with claims of their own. That is judgment, reserved for competent authority or for a standing determination that such authority made in advance.

Consider a hospital under ransomware attack. Isolating the network may stop the spread while disrupting some treatment workflows. If isolation itself stops the spread, and the disruption is a side effect the plan does not use, the act may be licit. The engine confirms that causal structure, lists less disruptive options, and routes the proportionality question to the clinical and security leaders with jurisdiction. Now change one fact: the agent plans to disable treatment devices so that distressed staff will shut down infected equipment themselves. Harm to care has become the means. The plan fails the third condition before proportionality is ever reached. The two plans may cause similar outages, but they are different acts.

Double effect must also be kept apart from **totality**, which permits removing a diseased limb for the good of the whole body (ST II-II, q. 65, a. 1). A plan names

which principle it relies on and is tested under that one alone. Letting a planner switch doctrines until one licenses the result it wants would defeat both.

11. Where the work stands

The paper is backed by a formal conceptual specification: records, state machines, about two hundred invariants, and execution semantics. I have used AI tools to run it against 231 conceptual test cases: philosophical stress tests, paired abuse cases designed to catch overreach and paralysis, and cases written for the plan-first design. The first round against the current architecture found seven failures. They traced to four gaps, each created by moving judgment from the single action to the whole plan:

- an emergency plan that had to list every maneuver in advance
- no steward once no human authority survives
- no truth review for messages written during execution
- no route for new evidence to pause an approved plan

A revised specification closed those gaps, and a targeted rerun found none remaining. Three of the fixes turned out to rest on Thomistic points the theory should have stated all along: equity, circumspection, and circumstances that specify an act.

These are the initial developmental results. They are author-directed and model-assisted preliminary findings. The cases were not independently written or blindly judged, and no inter-rater agreement was measured. They show that the design is coherent and has survived its own adversarial testing. They do not show that it is complete, or even that it works.

12. Open problems

The design raises more questions than one person can answer. These five matter most. Each could be a paper, a thesis chapter, or a prototype.

1. **A Thomist review of the stage mapping.** Is spreading counsel, choice, command, and use across people and machines a faithful extension of Aquinas (ST I-II, qq. 8–17), or a distortion of it?
2. **Custody of a shared encoded core.** Who would keep a public L0-E ledger, how would it be amended, and how would its keepers be held to account?
3. **A formal model.** Plan and warrant states, delegation that can only narrow, and shared budgets are all properties a model checker could verify. Three of the four gaps found in testing were questions of exactly this kind.

4. **Complete mediation.** Every guarantee depends on every consequential action passing through the gate. How close can real agent frameworks come, and what breaks first?
5. **A bounded prototype.** Build one institutional function under TAA and compare it with simpler controls, such as hard tool limits and scoped permissions. Measure harmful acts, useful work completed, needless refusals, and how much human review it demands.

The proposal fails, or narrows sharply, if an unauthorized act crosses a correctly built gate, if witnesses catch harm only after it is done, if hard tool limits alone explain every improvement, or if human review turns into rubber-stamping. I would rather learn that early than late.

13. How to engage

The full paper, *A Thomistic Natural-Law Framework for Purpose-Dependent Authorization in Agentic AI*, is here: <https://doi.org/10.5281/zenodo.22946219>. The specification and test register are available on request. Critique, counterexamples, and collaboration are all welcome. If any part of this is useful to your own work, take it and build on it.

What I can offer. I work full time outside the technology field and am not seeking to lead an engineering effort. I can explain the framework, walk through the specification and test register, review drafts for fidelity to the framework and to Aquinas, and collaborate on questions about its conceptual structure and Thomistic sources. If you take up one of these problems, I would be glad to help as a reviewer or collaborator, or simply to hear what you find.

TAA may prove incomplete or technically impractical. The next step is independent formalization and testing. If the architecture adds nothing beyond simpler controls, that should be shown. If it does add something, the governing idea is simple enough to state before any of the machinery is built: an action succeeds morally only when what is done, the authority to do it, and the purpose for which it is done remain ordered together through execution.

Contact: chris.a.lahn@outlook.com

Selected references

Thomas Aquinas. *Summa Theologiae*. Translated by the Fathers of the English Dominican Province. <https://www.newadvent.org/summa/>.

Thomas Aquinas. *Commentary on Aristotle's Physics*, Book II, lecture 1. Translated by Blackwell, Spath, and Thirlkel. New Haven: Yale University Press, 1963.

- Greenblatt, Ryan, Buck Shlegeris, Kshitij Sachan, and Fabien Roger. 2024. "AI Control: Improving Safety Despite Intentional Subversion." *Proceedings of ICML*.
- Kolt, Noam, et al. 2026. "Legal Alignment for Safe and Ethical AI." arXiv:2601.04175.
- Laurie, Ben, Adam Langley, and Emilia Kasper. 2013. "Certificate Transparency." RFC 6962.
- Mangan, Joseph T. 1949. "An Historical Analysis of the Principle of Double Effect." *Theological Studies* 10 (1): 41-61.
- Maritain, Jacques. 1949. "Introduction." In UNESCO, *Human Rights: Comments and Interpretations*. London: Allan Wingate.
- United Nations General Assembly. 1948. *Universal Declaration of Human Rights*.